STROBE Statement—checklist of items that should be included in reports of cohort studies

Discrimination transports, calibration does not: calibration drift in severity scoring across 187 intensive-care units

Guideline	STROBE 2007 (cohort studies); RECORD 2015 extension in Part B
Study design	Retrospective secondary analysis of two routinely collected critical-care databases
Journal	Scientific Reports
Items	22 STROBE items (34 sub-items) + 13 RECORD items
Section anchor	Heading names are stable across pagination; fill page numbers from the final typeset file

Part A — STROBE (cohort studies)

Item No	Recommendation	Section	Page No.	Relevant text from manuscript
1a	Indicate the study's design with a commonly used term in the title or the abstract.	Abstract, sentence 1		"In a secondary analysis of eICU-CRD (30,796 patients aged ≥65 years from 187 hospitals) we examined whether such scores discriminate and calibrate alike across centres."
1b	Provide in the abstract an informative and balanced summary of what was done and what was found.	Abstract (200/200 words, unstructured)		Reports the main estimate ("pooled AUC 0.732 … between-centre I^2 was 75.4% and the observed-to-expected mortality ratio ranged from 0.69 to 1.54, a 2.2-fold spread"), the recalibration penalty ("with 25 local patients it performed worse than no recalibration (success rate 15.5% versus 21.4%)"), and the shrinkage result ("outperforming naive recalibration at 25 patients (24.3% versus 15.5%; two-sided $P = 3.7 \times 10^{-112}$)"). Two negative findings are stated.
2	Explain the scientific background and rationale for the investigation being reported.	Introduction ¶1–3		"Calibration — whether the predicted probability of death matches the observed probability — is the property that determines what a risk estimate actually means at the bedside …"
3	State specific objectives, including any prespecified hypotheses.	Introduction, final ¶		"We quantified between-centre variation in calibration for three established severity scores, tested how much of it measured case-mix could explain, decomposed it into components attributable to centre size and to calendar year, translated it into consequences at four decision thresholds, and evaluated local recalibration as a function of the number of locally available patients — together with an empirical-Bayes shrinkage alternative."
4	Present key elements of study design early in the paper.	Methods → Study design and setting		"We performed a retrospective secondary analysis of two publicly available, de-identified, multicentre critical-care databases: the eICU Collaborative Research Database (eICU-CRD v2.0) and MIMIC-IV v3.1."
5	Describe the setting, locations, and relevant dates, including periods of recruitment, exposure, follow-up, and data collection.	Methods → Data sources		"eICU-CRD v2.0 contains data from 208 hospitals in the United States for the period 2014–2015, comprising 200,859 intensive-care unit (ICU) stays." … "MIMIC-IV v3.1 contains data from Beth Israel Deaconess Medical Center, a single academic medical centre, comprising 94,458 ICU stays."
6a	Give the eligibility criteria, and the sources and methods of selection of participants. Describe methods of follow-up.	Methods → Cohort construction; Fig. 1		"Of 200,859 unit stays across 208 hospitals, we retained the first ICU stay per patient (139,367), restricted to patients aged ≥ 65 years (71,734) with an ICU length of stay ≥ 24 h (50,138) and admitted to a medical ICU or medical–surgical ICU (31,307)." Centre-level eligibility: "a hospital had to contribute ≥ 100 patients and ≥ 10 in-hospital deaths … 103 of 187 hospitals met it."
6b	For matched studies, give matching criteria and the number of exposed and unexposed.	N/A		Not a matched design. No matching was performed; all eligible patients were included (see 6a).
7	Clearly define all outcomes, exposures, predictors, potential confounders, and effect modifiers. Give diagnostic criteria, if applicable.	Methods → Severity scores; → Centre-level calibration measures		"In-hospital death is a discharge status recorded in the source databases rather than a derived outcome, and no case-finding algorithm was applied to it." The two centre-level quantities are defined in full: "we solved for a_h such that the mean of expit(logit(p) + a_h) equals $\hat{y}$, by bisection over [−12, 12] to a tolerance of 10^{-4}"; and the calibration slope

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8	For each variable of interest, give sources of data and details of methods of assessment (measurement). Describe comparability of assessment methods if there is more than one group.	Methods → Data sources; → Severity scores		"Three established severity scores were used exactly as supplied by the source extraction protocols … This study did not recompute the scores, and their derivation is documented in the source protocols."
9	Describe any efforts to address potential sources of bias.	Methods → Potential sources of bias		"Four sources were considered." Selection (24-h stay and medical-ICU restriction; 24-h deaths excluded), information (scores and outcome not re-adjudicated), leakage ("each hospital's shrinkage parameters were estimated from the remaining hospitals only"), and confounding of the between-centre comparison ("No causal comparison between hospitals is made or intended").
10	Explain how the study size was arrived at.	Methods → Study size and missing data		"No sample size calculation was performed. This is a secondary analysis of two existing databases, and the analysis used all patients meeting the eligibility criteria described above."
11	Explain how quantitative variables were handled in the analyses. If applicable, describe which groupings were chosen and why.	Methods → Quantitative variables and cut-points		"All three scores, age and comorbidity burden were analysed as continuous variables and standardised before fitting; they were not categorised. Cut-points used in this study were fixed in advance and are not data-driven …" (all thresholds listed and traced to the pre-specified plans).
12a	Describe all statistical methods, including those used to control for confounding.	Methods → Pooled models through → Statistical reporting		"For each score we fitted a pooled logistic regression across the full cohort, with the score standardised (z-score) before fitting." Plus: random-effects meta-analysis with DerSimonian–Laird τ^2, leave-one-centre-out cross-fitting, paired two-sided Wilcoxon signed-rank tests, and cluster bootstrap for pooled AUC confidence intervals.
12b	Describe any methods used to examine subgroups and interactions.	Methods → Centre size and clinical impact; → Case-mix ladder		"Hospitals were split into tertiles by sample size, and I^2 and τ^2 were recomputed within each tertile." Three nested case-mix models (M1–M3) were also compared. No interaction test was performed; subgroup analyses are descriptive.
12c	Explain how missing data were addressed.	Methods → Study size and missing data; Supplementary Table S1		"In the eICU-CRD cohort the three severity scores, age, sex, invasive ventilation and in-hospital death were complete in every patient (0% missing), so no imputation was performed. Charlson comorbidity index was missing in 3,560 of 30,796 patients (11.6%) in eICU-CRD … and in 3 of 8,579 patients (0.04%) in MIMIC-IV." Comorbidity enters only the descriptive M3, which uses complete cases.
12d	If applicable, explain how loss to follow-up was addressed.	N/A		The outcome is in-hospital death, taken from the discharge status recorded in the source database. There is no post-discharge follow-up and therefore no loss to follow-up.
12e	Describe any sensitivity analyses.	Methods → Between-centre heterogeneity; → Temporal stability; → Analysis framework		"Because DerSimonian–Laird predates current guidance, τ^2 was additionally re-estimated under three further frameworks as a pre-specified analysis (analysis K): Paule–Mandel, restricted maximum likelihood (REML), and a one-stage random-intercept multilevel logistic model …" The case-mix ladder (M1–M3) and a raw-scale (unstandardised) refit serve as further sensitivity analyses.
13a	Report the numbers of individuals at each stage of study — e.g. numbers potentially eligible, examined for eligibility, confirmed eligible, included in the study, completing follow-up, and analysed.	Results → Cohort and study setting; Fig. 1		"The eICU-CRD cohort comprised 30,796 patients from 187 hospitals, of whom 3,901 (12.7%) died in hospital; 103 hospitals met the pre-specified centre-level eligibility criterion of ≥ 100 patients and ≥ 10 in-hospital deaths and entered the centre-level analyses … The MIMIC-IV cohort comprised 8,579 patients from a single academic medical centre, of whom 1,654 (19.3%) died (Table 1; Supplementary Table S1)." All intermediate counts are in Fig. 1 and Methods.
13b	Give reasons for non-participation at each stage.	Fig. 1; Methods → Cohort construction		"We then excluded deaths within 24 h of ICU admission (n = 7), stays with an unknown hospital outcome status (n = 363), and stays in

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				which albumin, height and weight were all missing (n = 141).” Reasons are given for every exclusion step in both cohorts.
13c	Consider use of a flow diagram.	Fig. 1		“The step-wise inclusion and exclusion criteria are shown in Figure 1, and the counts reported there are taken from the extraction stage's own count reports … rather than re-derived, so that Figure 1 reflects the actual pipeline rather than a reconstruction.”
14a	Give characteristics of study participants (eg demographic, clinical, social) and information on exposures and potential confounders.	Table 1; Supplementary Table S1		Table 1 reports, by score and cohort, the number of patients and hospitals, age, sex, invasive ventilation, Charlson index, observed mortality and the pooled discrimination and calibration estimates. Supplementary Table S1 gives the full cohort characteristics with a missing-data row for every variable.
14b	Indicate the number of participants with missing data for each variable of interest.	Supplementary Table S1; Methods → Study size and missing data		“Charlson comorbidity index was missing in 3,560 of 30,796 patients (11.6%) in eICU-CRD (2,807 of 26,436, 10.6%, within the 103-centre analysis subset) and in 3 of 8,579 patients (0.04%) in MIMIC-IV.” All other analysis variables were 0% missing.
14c	Summarise follow-up time (eg, average and total amount).	Methods → Cohort construction; → Study size and missing data		The observation window is the index hospital admission; “In-hospital death is a discharge status recorded in the source databases”. ICU length of stay ≥ 24 h was an eligibility criterion, so no patient contributed less than one day of observation.
15	Report numbers of outcome events or summary measures over time.	Results → Cohort and study setting; Table 1; Supplementary Table S1		eICU-CRD: “3,901 (12.7%) died in hospital” of 30,796. MIMIC-IV: “1,654 (19.3%) died” of 8,579. Per-hospital event counts are the input to every centre-level estimate and are recoverable from the archived code.
16a	Give unadjusted estimates and, if applicable, confounder-adjusted estimates and their precision (eg, 95% confidence interval). Make clear which confounders were adjusted for and why they were included.	Results → Calibration varies widely across centres; Table 1; Fig. 3		“For SOFA, the pooled model discriminated moderately (AUC 0.732)” (95% CI 0.720–0.745 by cluster bootstrap, 1,000 replicates). Covariates enter only the case-mix ladder, where the reason is stated: “To test whether measured case-mix explains between-centre calibration drift”. Precision is reported for every estimate.
16b	Report category boundaries when continuous variables were categorised.	Methods → Quantitative variables and cut-points		“the centre-size tertile boundaries (hospitals split by sample size into three groups of 36, 33 and 34 hospitals, with ranges 100–181, 184–280 and 289–987 patients respectively)”. The four decision thresholds (0.10, 0.20, 0.30, 0.40) are also given. Scores and age were not categorised.
16c	If relevant, consider translating estimates of relative risk into absolute risk for a meaningful time period.	N/A		No relative-risk estimates are reported. The study reports absolute risks directly — observed mortality, predicted mortality, the observed-to-expected ratio, and mortality among patients flagged above each decision threshold.
17	Report other analyses done — eg analyses of subgroups and interactions, and sensitivity analyses.	Results → Measured case-mix; → Drift strengthens; → Calibration offset is not a stable attribute; → The between-centre variance depends on the estimation framework; Supplementary Fig. S1		Reports the case-mix ladder with its two-way decomposition (“1.9 points were therefore attributable to the change in analysis set and 3.4 points to the addition of the covariates themselves”), the size tertiles, the split-half temporal analysis, and τ^2 under four frameworks.
18	Summarise key results with reference to study objectives.	Discussion ¶1		“In 103 hospitals using the same scores and the same pipeline, a SOFA-based model discriminated consistently (per-centre AUC 0.649–0.812) while its calibration varied by more than twofold (observed-to-expected mortality 0.69–1.54; between-centre I^2 = 75.4%).”
19	Discuss limitations of the study, taking into account sources of potential bias or imprecision. Discuss both the direction and magnitude of any potential bias.	Discussion → Limitations		“First, both databases were collected for purposes other than this study … Misclassification in the underlying records … cannot be detected here, and its direction and magnitude could not be quantified.” Six limitations in total, each with its direction of effect where the direction is knowable.
20	Give a cautious overall interpretation of results considering objectives, limitations,	Discussion ¶1–6; Methods → Analysis framework and		Findings are set against prior calibration-drift and model-updating work, with the qualifying

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	multiplicity of analyses, results from similar studies, and other relevant evidence.	inferential status		rather than contradicting relation made explicit. Multiplicity is addressed directly: “no formal adjustment for multiple comparisons was applied”, with analyses assigned to pre-specified confirmatory or descriptive tiers and two unmet criteria reported as negative findings.
21	Discuss the generalisability (external validity) of the study results.	Discussion → Limitations, final paragraph		“the empirical-Bayes parameters were estimated in a single database, and although they transferred to a second, the extent to which they generalise to other health systems and other score families remains open.” The scope of inference is further bounded by the 2014–2015 eICU-CRD window.
22	Give the source of funding and the role of the funders for the present study and, if applicable, for the original study on which the present article is based.	Additional Information → Funding		“This research received no external funding.” No funder had any role in design, analysis, interpretation, or the decision to submit.

Part B — RECORD extension (routinely collected health data)

Item No	Recommendation	Section	Page No.	Relevant text from manuscript
R1.1	The type of data and the name of the database(s) used should be reported in the title or abstract.	Abstract		"In a secondary analysis of eICU-CRD …" and "… transferring to MIMIC-IV without refitting." Both databases are named in the abstract.
R1.2	The geographic region and timeframe within which the study took place should be reported in the title or abstract.	Abstract (not reported); Methods → Data sources		Not in the abstract, which is at its 200-word limit. Region and period are reported in Methods: United States, 2014–2015 (eICU-CRD); single US academic centre (MIMIC-IV). Author decision, recorded here.
R1.3	If the study used a linkage between databases, this should be described in the title or abstract.	N/A		No record linkage between databases. The two cohorts are analysed independently.
R6.1	The methods of selection of the study population using codes or algorithms should be reported.	Methods → Cohort construction; Fig. 1		Every inclusion and exclusion step is reported with its count and its criterion. No code list was required, because eligibility rests on age, ICU type, length of stay and recorded discharge status rather than on diagnostic coding.
R6.2	The validation of the codes or algorithms used to select the population should be referenced.	Methods → Validity of the variable definitions		"SOFA, OASIS and SAPS-II are established, previously published and externally validated scores, used here exactly as supplied by the source extraction protocols; their components and weights are those of the original publications and are not modified in this study."
R6.3	If the study involved linkage of databases, a flow diagram of the linkage should be provided.	N/A		No linkage; consequently no linkage flow diagram. Fig. 1 shows participant flow for each cohort separately.
R7.1	A complete list of codes and algorithms used to classify exposures, outcomes, confounders, and effect modifiers should be provided.	Methods → Validity of the variable definitions; → Centre-level calibration measures		"No code list or phenotype algorithm was developed for this study; where a variable required a derived definition, that definition is given above in full (the calibration offset and its standard error, and the calibration slope)." Both derived definitions are given as closed-form expressions.
R12.1	Describe the extent to which the investigators had access to the database population.	Methods → Data sources		"All database access was read-only; no script written for this study modified a database." Access was to the complete credentialed databases, not to a restricted extract.
R12.2	Describe the methods used to clean the data.	Methods → Cohort construction; → Study size and missing data		Cleaning is confined to the reported eligibility filters; no value was corrected, imputed or winsorised. Missingness is quantified for every variable rather than handled by a cleaning rule.
R12.3	Describe the methods used to link databases, and the quality of the linkage.	N/A		No linkage was performed, so linkage quality does not apply.
R13.1	Describe the selection of participants, including data-quality and completeness filters.	Methods → Cohort construction; Fig. 1		Includes the data-availability filters: "stays with an unknown hospital outcome status (n = 363)" and "stays in which albumin, height and weight were all missing (n = 141)", each with its count and its reason.
R19.1	Discuss the implications of using data that were not created to answer the research question.	Discussion → Limitations, first limitation		"Misclassification in the underlying records — in the severity scores, in comorbidity coding and in the recorded discharge status — cannot be detected here, and its direction and magnitude could not be quantified. And because eICU-CRD covers only 2014 and 2015, we could not assess how eligibility for the cohort, or the coding practices behind it, changed over time."
R22.1	Authors should provide information on how to access the study protocol, raw data, and programming code.	Data Availability; Methods → Software, seed and reproducibility		Analysis plans and their timestamps are reproduced in the Supplementary Note; code and derived results are archived with a DOI; raw data are available from PhysioNet under the original data use agreements.

Notes on items recorded as N/A

Item	Reason
6b	Not a matched design; no matching was performed.
12d	Outcome is in-hospital death recorded at discharge; no post-discharge follow-up, hence no loss to follow-up.
16c	No relative-risk estimates; absolute risks and observed-to-expected ratios are reported directly.
R1.3 / R6.3 / R12.3	The two databases were analysed independently; no record linkage was performed.